

Using LLMs to Model Arguments in U.S. Supreme Court Briefs: Preliminary Report ^{*}

Heng Zheng¹, Dexter Williams², and Bertram Ludäscher³

¹ School of Information Science, University of Kentucky, USA hengzheng@uky.edu

² The Information School, University of Wisconsin-Madison, USA djwilliams22@wisc.edu

³ School of Information Sciences, University of Illinois, Urbana-Champaign, USA
ludaesch@illinois.edu

Abstract. Supreme Court briefs can impact the decision-making of Supreme Court cases. We show how to represent the briefs for a U.S. Supreme Court case as a formal argument model that relies on the briefs’ existing structure. We explore how to generate argument models with large language models (LLMs) by using them to identify attacks between arguments.

Keywords: U.S. Supreme Court · large language model · computational argumentation · argument mining

1 Introduction

Supreme Court briefs can impact the decision-making of Supreme Court cases [4]. Some U.S. Supreme Court cases have received extensive attention outside of law, such as the *Roe v. Wade*⁴ case being overturned, receiving over 70 million discussions on Twitter (now X) in about one year (January 2022 to January 2023) [3]. Helping the public understand how the Supreme Court makes decisions to mitigate misunderstandings of the legal decision-making process is important [13].

Computational argumentation methods can assist in analyzing Supreme Court briefs by identifying the arguments presented in the briefs and mapping the relationships between them. Here, we investigate:

1. How can we model U.S. Supreme Court briefs with computational argumentation?
2. Can we use LLMs to automatically generate the models?

We show how to model the Supreme Court briefs as argument models and explore the potential for LLMs to assist in the modeling process. This can assist non-experts in comprehending complex legal knowledge in the briefs, such as how different stakeholders (petitioners, respondents, and amici curiae) argue for their stances and against other parties.

^{*} This paper is an extended version of our JURIX 2025 poster paper: “Modeling U.S. Supreme Court Briefs with Computational Argumentation”.

⁴ *Roe v. Wade*, 410 U.S. 113 (1973).

2 Related Work

Research on the U.S. Supreme Court cases focuses on the development of rules over time [9] using the Fourth Amendment’s automobile exception, and oral arguments analysis [2,1]. AI techniques can also improve Supreme Court briefs analysis to advance understanding of what arguments from involved parties were considered when deciding the case, such as using machine learning to classify the ideology of briefs [5].

Transformer-based language models have been utilized to analyze legal documents, combining computational argumentation approaches by incorporating large language models with ASPIC+ to generate arguments [11] and using argument schemes with transformer-based natural language models to mine arguments in legal documents [8].

Using AI techniques, especially LLMs, to analyze Supreme Court briefs requires evaluating LLMs’ performance in understanding the arguments in the brief. AI has shown its ability in argument summarization and completion using a benchmarking dataset of U.S. Supreme Court briefs [15]. AI’s performance in identifying relationships between arguments varied: LLM-based approaches show good performance in distinguishing support and attack relationships [6]. However, in [6], no datasets from the legal domain were used for experimentation. In contrast, Graph Neural Network-based approaches did not perform well in identifying undercutting attacks [16]. Further investigation is still needed to determine whether LLMs can effectively identify attack relationships between arguments from opposite Supreme Court case briefs.

3 Modeling the U.S. Supreme Court Briefs with Formal Argumentation

We chose the *Acheson Hotels, LLC v. Laufer*⁵ case as an example to illustrate how we model the Supreme Court briefs. Both the petitioner and the respondent received amicus briefs that supported them. We focused on the briefs filed during the merits stage instead of the certiorari stage, because not only do the judges use these briefs to help make their final judgment, but also because the petitioner and the respondent may not be opposed to having the case heard in the cert stage. Here is a short introduction to the case:

“Deborah Laufer, a prolific litigant with physical disabilities and vision impairments, sued Acheson Hotels for failing to publish information about their accessibility on their website, which is required under the Americans with Disabilities Act (ADA).

The district court dismissed the lawsuit, finding that Laufer lacked standing to sue because had no plans to visit the hotel and thus suffered no injury as a result of the lack of information on the website. The U.S. Court of Appeals for the First Circuit reversed, concluding that Laufer’s lack of intent to book a room at the hotel operated by Acheson does not negate the fact of injury.”⁶

We selected the following briefs from the petitioner, the respondent, and one amici curiae that supports the respondent:

⁵ *Acheson Hotels, LLC v. Laufer*, No. 22–429 (U.S. Dec. 5, 2023).

⁶ *Facts of the case* from *oyez.org*: <https://www.oyez.org/cases/2023/22-429>

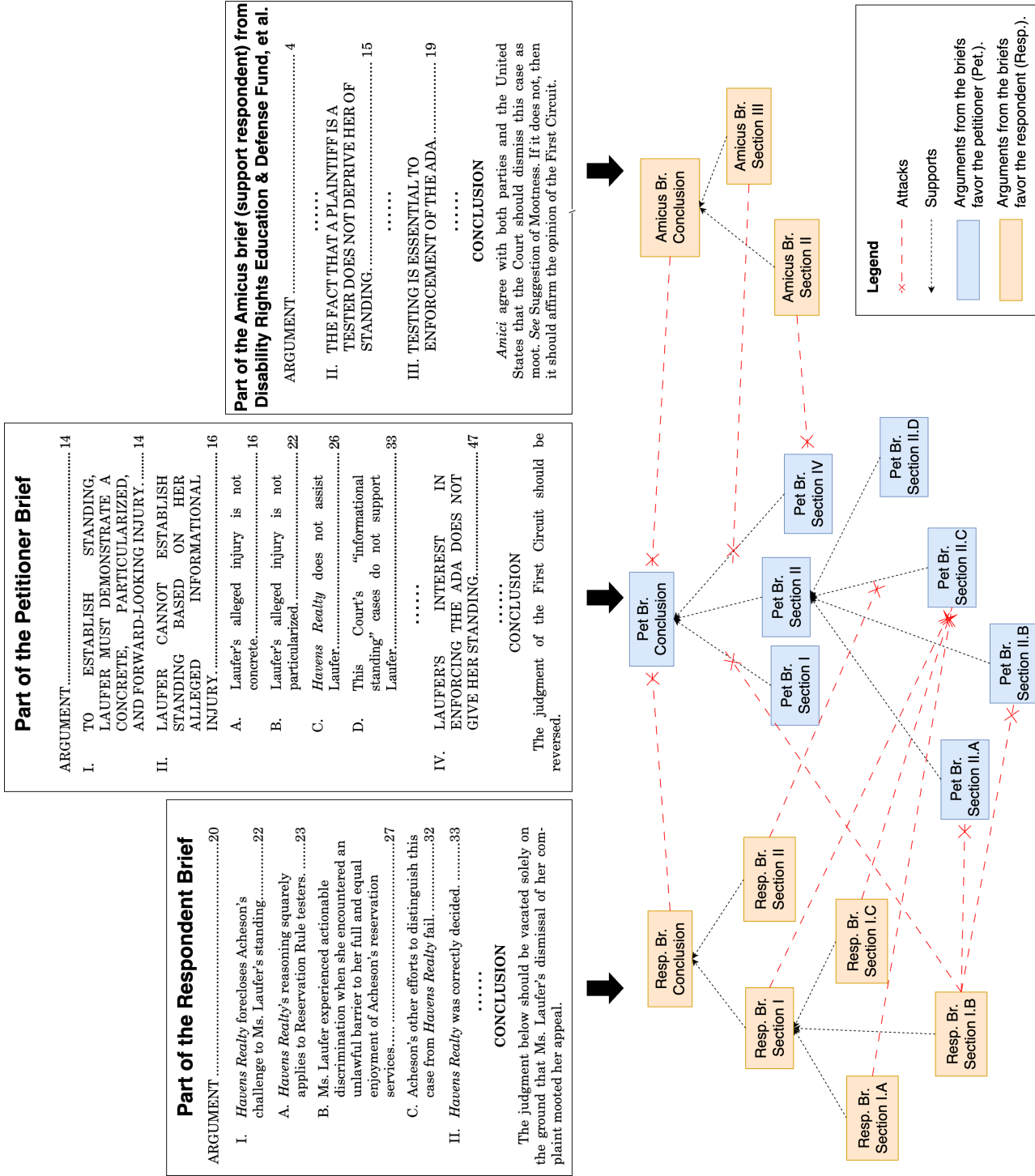


Fig. 1: Modeling excerpts from the petitioner brief (left), the respondent brief (middle), and an amicus brief (right) that supports the respondent as an argument model

1. The petitioner’s (Acheson Hotels) brief⁷ filed at the beginning of the merits stage.
2. The respondent’s (Laufer) brief⁸ that was filed in response to the petitioner’s brief.
3. An amicus brief⁹ filed by *Disability Rights Education & Defense Fund et al.* that supports the respondent and against the petitioner. *Disability Rights Education & Defense Fund* is an advocacy group for individuals with disabilities.

The briefs follow a strict structure [12]. Figure 1 shows the table of contents of sample briefs and how we model the *Acheson Hotels* case’s briefs in an argument model. We model each brief as an argument. The subsections under the ARGUMENT section are premises that support the conclusion (content in the CONCLUSION section). Here, we deem each subsection to *support* the section and/or the conclusion independently, and the inference from the subsection (as a premise) to the section/conclusion is *defeasible*. For example, the part of the petitioner’s brief shown in Figure 1 has Sections I, II, and IV that support the conclusion, and Section II is further supported by Sections II.A, II.B, II.C, and II.D. Arguments, such as the one from “Pet Br. Section II.C” to “Pet Br. Section II”, are defeasible, because other arguments can attack them.

We consider attacks between arguments if they respect the temporal filing order, thus capturing how stakeholders argue in response to other parties. Therefore, attacks only exist from a later-filed brief to an earlier-filed brief, and the two briefs should not favor the same sides. The timeline for filing briefs during the merits stage is as follows (see Rules 24, 25, and 37 in [14]): The petitioner first files a brief to argue on the merits. Then, the respondent (the opposite party of the petitioner) files a brief to respond to the petitioner’s argument. After that, the petitioner may reply to the respondent’s argument. Amici curiae that support each party can also file briefs to present their arguments during the merits stage. By keeping the order in which attacks were made, we model the decision-making process (briefings can attack earlier but not future arguments).

Attacks are classified as *undermining*, *rebutting*, or *undercutting*: Undermining is an attack on a premise, rebutting is an attack on a conclusion, while undercutting is an attack on the inference relation between premise(s) and conclusion [10]).

Arguments in the respondent’s brief and the amicus brief in Figure 1 attack the argument in the petitioner’s brief. The respondent’s brief concludes that the First Circuit’s judgment should be “vacated,” which is a rebuttal of the petitioner’s brief’s conclusion: the judgment should be “reversed.”

Respondent brief’s Section I: “*Havens Realty*¹⁰ forecloses Acheson’s challenge to Ms. Laufer’s standing” is an undermining attack of the petitioner brief’s Section II.C: “*Havens Realty* does not assist Laufer.” Because Section II.C is modeled as a premise in the argument of the petitioner’s brief. The *Havens Realty* case is a key precedent in the *Acheson Hotels* case. “*Havens Realty* forecloses Acheson’s challenge to Ms. Laufer’s standing” directly against the petitioner’s point on *Havens Realty does not assist Laufer*.

In contrast, Section II in the respondent’s brief (“*Havens Realty* was correctly decided”) is an undercutting attack of the petitioner’s Section II.C. Because it does not

⁷ Brief of petitioner Acheson Hotels, LLC

⁸ Brief of respondent Deborah Laufer

⁹ Brief amici curiae of *Disability Rights Education & Defense Fund*

¹⁰ *Havens Realty Corp. v. Coleman*, 455 U.S. 363 (1982).

directly argue whether *Havens Realty* supports Laufer (respondent) or not, instead it argues whether the *Havens Realty* was correctly decided.

4 Using LLMs to Generate the Argument Model

Constructing an argument model for the briefs filed in the *Acheson Hotels* case is time-consuming. LLMs can help build up the model more efficiently.

To use AI for the argument model construction, we attempted to use LLMs to identify attacks between the sections from opposing briefs (e.g., respondent vs petitioner). This allows us to generate structured argument models composed of:

- Arguments with premises as the summary of the content in each subsection under the ARGUMENT section, and conclusions as the content of the CONCLUSION section;
- Attacks between the arguments.

For attack-relation mining, Mixtral-8x7B and Mistral7B have shown strong performance on relation-based argument mining (RbAM) tasks in non-legal datasets [6]. However, RbAM has not yet been applied to U.S. Supreme Court briefs. Given its availability as an open-weight model and its prior use in [6], we chose Mistral7B to evaluate Task 2, reusing the prompt from [6] with slight adaptation:

Prompt for attack relation identification

In this task, you will be given two arguments and your goal is to classify the relation between them as either “attack” or “no” based on the definitions below. Attack: It is an argument that contradicts or opposes the parent argument. No: It is an argument that has no relation to the parent argument. Answer with one word.

Arg1: Parent Argument(parent)

Arg2: Child Argument(child)

Relation:

Our annotation includes the petitioner’s brief, the respondent’s brief, the reply brief of the petitioner¹¹, and the amicus brief filed by *Disability Rights Education & Defense Fund, et al.* Our annotation is based on the section headings of the briefs under the ARGUMENT section, because these can be viewed as the summary of the whole sections. We compared the section headings and checked whether they attack each other. The attacks can be further characterized as undermining or rebutting, depending on whether they are premises or conclusions. Here annotations are used to show the potential of LLMs for identifying attack relations, not for evaluation purposes. We also have not identified undercutting attacks between arguments in our initial experiments.

While a lack of annotated data currently makes meaningful evaluation challenging, the initial output shows Mistral7B’s performance varied. As an example, Mistral7B

¹¹ [Reply of petitioner Acheson Hotels, LLC](#)

successfully identified the conflict between “*Havens Realty* forecloses Acheson’s challenge to Ms. Laufer’s standing” from the respondent brief and “*Havens Realty* does not assist Laufer” from the petitioner brief. However, Mistral7B missed other undermining attacks from the respondent’s brief to “*Havens Realty* does not assist Laufer” from the petitioner brief, such as “*Havens Realty*’s reasoning squarely applies to Reservation Rule testers,” where Mistral7B failed to connect Laufer with “Reservation Rule testers.”

5 Discussion and Future Work

Based on the first author’s experience annotating the briefs in the *Acheson Hotels* case, a full expert analysis of arguments in U.S. Supreme Court briefs can be very time-consuming. The structure of the U.S. Supreme Court briefs allows us to model the briefs as an argument model. It can be difficult for non-legal experts to determine whether the existing headings in the briefs conflict with headings from the opposite briefs. For example, the section heading “*This [Supreme] Court’s ‘information standing’ cases do not support Laufer*” from the petitioner’s brief does not indicate what the ‘information standing’ cases are, and thus make it hard for non-legal experts to determine whether the *Havens Realty* case mentioned by the respondent is one of them.

The section headings in the briefs may also be too abstract for LLMs to correctly identify all attack relations. In our initial attempt, Mistral7B can identify the conflict between “*Havens Realty* forecloses Acheson’s challenge to Ms. Laufer’s standing” from the respondent and “*Havens Realty* does not assist Laufer” from the petitioner, but failed to connect Laufer with “Reservation Rule testers”. Therefore, more context will be helpful for LLMs to conduct relation-based argument mining, such as a more detailed summarization by LLMs of the subsections under the ARGUMENT section of the briefs.

How to use LLM to identify undercutting attacks between arguments still needs further investigation. Based on similar tasks performed on a dataset of social media [16], identifying undercutting attacks can be a challenge.

Future work is needed to properly evaluate LLMs’ performance. To this end, we will create a legal expert-annotated corpus of argument relations in Supreme Court briefs for the LLM-based attack identification, then use the dataset to evaluate the performance of different LLMs for attack relation mining on Supreme Court briefs and compare the performance of different prompts. We will experiment with different LLMs and see if incorporating other techniques can help identify the attacks, such as factor extraction [7]. We will evaluate whether the relation mining performance depends on the level of detail in the argument summarization. We also plan to explore methods to present insights to the public based on the argument models. While our initial model was inspired by ASPIC+ [10], subsequent work will develop a more formal, structured model of Supreme Court briefs. This will allow us to analyze the briefs using the computational semantics of (structured) argumentation frameworks.

6 Summary and Conclusions

Through a preliminary study, we have investigated how to combine the strengths of LLMs and computational argumentation to assist non-legal experts in analyzing the U.S. Supreme Court briefs. We have modeled the briefs as argument models: The sections in a brief can form the premises and conclusions of an argument, and attacks can be derived between the briefs that do not favor the same party. We have explored the use of LLMs to identify the attacks between the arguments in order to generate argument models more efficiently. While a lack of annotated data currently makes meaningful evaluation challenging, we find the initial results promising. Future work is needed to evaluate LLMs' performance on attack relation identification systematically.

References

1. Al-Abdulkarim, L., Atkinson, K., Bench-Capon, T.: From Oral Hearing to Opinion in the U.S. Supreme Court. In: *Legal Knowledge and Information Systems*. pp. 1–10. IOS Press (2013). <https://doi.org/10.3233/978-1-61499-359-9-1>
2. Ashley, K., Pinkwart, N., Lynch, C., Alevén, V.: Learning by diagramming Supreme Court oral arguments. In: *Proceedings of the 11th international conference on Artificial intelligence and law*. pp. 271–275. ICAIL '07, Association for Computing Machinery, New York, NY, USA (Jun 2007). <https://doi.org/10.1145/1276318.1276370>
3. Chang, R.C., Rao, A., Zhong, Q., Wojcieszak, M., Lerman, K.: #RoeOverturned: Twitter Dataset on the Abortion Rights Controversy. *Proceedings of the International AAAI Conference on Web and Social Media* **17**, 997–1005 (Jun 2023). <https://doi.org/10.1609/icwsm.v17i1.22207>
4. Corley, P.C.: The Supreme Court and Opinion Content: The Influence of Parties' Briefs. *Political Research Quarterly* **61**(3), 468–478 (Sep 2008). <https://doi.org/10.1177/1065912907306474>
5. Evans, M., McIntosh, W., Lin, J., Cates, C.: Recounting the Courts? Applying Automated Content Analysis to Enhance Empirical Legal Research. *Journal of Empirical Legal Studies* **4**(4), 1007–1039 (2007). <https://doi.org/10.1111/j.1740-1461.2007.00113.x>
6. Gorur, D., Rago, A., Toni, F.: Can Large Language Models perform Relation-based Argument Mining? In: Rambow, O., Wanner, L., Apidianaki, M., Al-Khalifa, H., Eugenio, B.D., Schockaert, S. (eds.) *Proceedings of the 31st International Conference on Computational Linguistics*. pp. 8518–8534. Association for Computational Linguistics, Abu Dhabi, UAE (Jan 2025), <https://aclanthology.org/2025.coling-main.569/>
7. Gray, M., Savelka, J., Oliver, W., Ashley, K.: Using LLMs to Discover Legal Factors. In: *Legal Knowledge and Information Systems*. pp. 60–71. IOS Press (2024). <https://doi.org/10.3233/FAIA241234>
8. Habernal, I., Faber, D., Recchia, N., Bretthauer, S., Gurevych, I., Spiecker genannt Döhmann, I., Burchard, C.: Mining Legal Arguments in Court Decisions. *Artificial Intelligence and Law* **32**(3), 1–38 (Sep 2024). <https://doi.org/10.1007/s10506-023-09361-y>
9. Henderson, J., Bench-Capon, T.: Describing the Development of Case Law. In: *Proceedings of the Seventeenth International Conference on Artificial Intelligence and Law*. pp. 32–41. ICAIL '19, Association for Computing Machinery, New York, NY, USA (Jun 2019). <https://doi.org/10.1145/3322640.3326697>
10. Modgil, S., Prakken, H.: The ASPIC+ Framework for Structured Argumentation: A Tutorial. *Argument & Computation* **5**(1), 31–62 (Jan 2014). <https://doi.org/10.1080/19462166.2013.869766>

11. Park, S., Choi, A., Park, R.: Objection, Your Honor!: An LLM-Driven Approach for Generating Korean Criminal Case Counterarguments. *Artificial Intelligence and Law* (Feb 2025). <https://doi.org/10.1007/s10506-025-09432-2>
12. Schweitzer, D.: U.S. Supreme Court Brief Writing Style-Guide Developments. *Journal of Appellate Practice and Process* **19**(1), 129–156 (2018), <https://heinonline.org/HOL/P?h=hein.journals/japp19&i=137>
13. Sullivan, B., Feldbrin, R.: The Supreme Court and the People: Communicating Decisions to the Public. *University of Pennsylvania Journal of Constitutional Law* **24**(1), 1–92 (2022), <https://heinonline.org/HOL/P?h=hein.journals/upjcl24&i=2>
14. Supreme Court of the United States: Rules of the Supreme Court of the United States. Tech. rep., Supreme Court of the United States (2023), <https://www.supremecourt.gov/filingandrules/2023RulesoftheCourt.pdf>
15. Woo, J., Hashemi Chaleshtori, F., Marasovic, A., Marino, K.: BriefMe: A Legal NLP Benchmark for Assisting with Legal Briefs. In: Che, W., Nabende, J., Shutova, E., Pilehvar, M.T. (eds.) *Findings of the Association for Computational Linguistics: ACL 2025*. pp. 13139–13190. Association for Computational Linguistics, Vienna, Austria (Jul 2025). <https://doi.org/10.18653/v1/2025.findings-acl.681>
16. Ye, Y., Teufel, S.: Computational modelling of undercuts in real-world arguments. In: Ajjour, Y., Bar-Haim, R., El Baff, R., Liu, Z., Skitalinskaya, G. (eds.) *Proceedings of the 11th Workshop on Argument Mining (ArgMining 2024)*. pp. 59–68. Association for Computational Linguistics, Bangkok, Thailand (Aug 2024). <https://doi.org/10.18653/v1/2024.argmining-1.6>